Iteration and simulation

Warm-up question

Problem: 10 people are at a party, and all of them are wearing hats. They each place their hat in a pile; when they leave, they choose a hat at random. What is the probability at least one person selected the correct hat?

Question: Work with your neighbor to discuss the following question:

- Without calculating probabilities, how could you design an experiment to estimate this probability?

Designing an experiment

Step 1: representing the hats

```r
hats <- 1:10

hats
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

```r
hats[3]
```

```
[1] 3
```

- `hats` is a **vector**, containing the numbers 1 to 10
- entries in a vector are accessed by their index

Step 2: everyone draws a random hat

```r
hats <- 1:10
randomized_hats <- sample(hats, size = 10, replace = FALSE)

hats
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

```r
randomized_hats
```

```
 [1]  8  3  2  7  1  9  5  4 10  6
```

- The `sample` function creates a random sample from a vector
- How many people selected their original hat?

Step 3: check who got their original hat

```r
hats <- 1:10
randomized_hats <- sample(hats, size = 10, replace = FALSE)
```

```r
hats
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

```r
randomized_hats
```

```
 [1]  8  3  2  7  1  9  5  4 10  6
```

```r
hats == randomized_hats
```

```
 [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

```r
# TRUE is 1, FALSE is 0
sum(hats == randomized_hats)
```

```
[1] 0
```

```r
# did at least one person get their hat?
sum(hats == randomized_hats) > 0
```

```
[1] FALSE
```

Code so far

```
hats <- 1:10
randomized_hats <- sample(hats, size = 10, replace = FALSE)
```

```
sum(hats == randomized_hats) > 0
```

```
[1] FALSE
```

- Is this a good estimate of the probability?

Step 4: iteration

A `for` loop repeats code many times:

```
nsim <- 10000 # number of simulations
for(i in 1:nsim){


}
```

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A `for` loop repeats code many times:

```
nsim <- 10000 # number of simulations
hats <- 1:10
results <- rep(NA, nsim) # vector to store results

for(i in 1:nsim){
  randomized_hats <- sample(hats, size = 10, replace = FALSE)
  results[i] <- sum(hats == randomized_hats) > 0
}

head(results)
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE
```

Step 4: iteration

A `for` loop repeats code many times:

```r
nsim <- 10000 # number of simulations
hats <- 1:10
results <- rep(NA, nsim) # vector to store results

for(i in 1:nsim){
  randomized_hats <- sample(hats, size = 10, replace = FALSE)
  results[i] <- sum(hats == randomized_hats) > 0
}

mean(results)
```

```
[1] 0.6262
```

- What if I wanted to repeat the simulation, with a different number of people?

Removing magic numbers

Without magic numbers:

```
nsim <- 10000 # number of simulations
n_people <- 10 # number of people
hats <- 1:n_people
results <- rep(NA, nsim) # vector to store results

for(i in 1:nsim){
  randomized_hats <- sample(hats,
                            size = n_people,
                            replace = FALSE)
  results[i] <- sum(hats ==
                      randomized_hats) > 0
}

mean(results)
```

```
[1] 0.6344
```

- Why did I get different results?

Final code

```r
set.seed(3) # set a seed for reproducibility

n_people <- 10 # number of people at the party
hats <- 1:n_people # numbered hats
nsim <- 10000 # number of simulations
results <- rep(NA, nsim) # vector to store the results

for(i in 1:nsim){
  # hats are randomly assigned to each person
  randomized_hats <- sample(hats, n_people, replace = F)

  # did at least one person get their hat back?
  results[i] <- sum(randomized_hats == hats) > 0
}

mean(results)
```

Class activity

- Work with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)